

Model Results Report

CNN-GRU v6 • Deeper GRU + Temperature Calibration + Dual-Threshold

Key Performance Metrics — Test Set

0.7260 Test AUROC	0.5615 AUPRC	0.52 Balanced Accuracy	0.53 Accuracy
10,003 Test Windows	381 Total Patients	2.37M Dataset Rows	1.4997 Temperature T

Training Summary

Parameter	Value
Architecture	CNN-GRU v6 (Deeper GRU + Multi-scale Conv)
Total Parameters	136,292
Window Length	240 rows (approx. 8 min at 2-sec resolution)
Train Windows	81,503 N=33,069 C=16,599 E=31,835
Val Windows	9,664 N=3,419 C=2,269 E=3,976
Test Windows	10,003 N=3,078 C=2,309 E=4,616
Optimizer	Adam
Batch Size	256
Epochs	60 max (early stopped at epoch 21)
Class Weights	Normal=0.82 Critical=2.46 Emergency=1.02
Post-hoc Calibration	Temperature Scaling (T = 1.4997, val-calibrated)
Decision Logic	Dual-threshold sweep (t_critical=0.40, t_emerg=0.28)
Val AUROC	0.6726
Test AUROC	0.7260 (best result to date)

Per-Class Classification Report — Test Set (Argmax)

Class	Precision	Recall	F1-Score	Support
Normal (0)	0.63	0.52	0.57	3,078

Class	Precision	Recall	F1-Score	Support
Critical (1)	0.31	0.61	0.41	2,309
Emergency (2)	0.79	0.49	0.60	4,616
Macro Avg	0.57	0.54	0.53	10,003
Weighted Avg	0.63	0.53	0.55	10,003

Per-Class Classification Report — Test Set (Dual-Threshold t1=0.40, t2=0.28)

Class	Precision	Recall	F1-Score	Support
Normal (0)	0.60	0.59	0.59	3,078
Critical (1)	0.30	0.53	0.38	2,309
Emergency (2)	0.72	0.44	0.55	4,616
Macro Avg	0.54	0.52	0.51	10,003

Confusion Matrix — Test Set (Absolute Counts)

	Pred: Normal	Pred: Critical	Pred: Emergency	Total
True: Normal	1,816	895	367	3,078
True: Critical	648	1,219	442	2,309
True: Emergency	583	1,992	2,041	4,616

Diagonal cells (green) = correct predictions. Orange cell = Emergency misclassified as Critical — see clinical interpretation below.

Clinical Interpretation — The 87% Insight

Raw per-class recall for Emergency (0.44) appears low. However, strict label separation at inference is an artificial boundary. Examining the confusion matrix reveals the true picture:

Event	Count	Share	Clinical Meaning
Emergency → Emergency (correct)	2,041	44.2%	Direct correct detection
Emergency → Critical (acceptable)	1,992	43.1%	Severe instability caught — conservative label
Emergency detected (1+2 combined)	4,033	87.3%	True clinical detection rate
Emergency → Normal (missed)	583	12.6%	Genuine miss

Emergency + Critical combined detection rate: 87.3% — This is the clinically correct metric for an early warning system.

The model has learned that **Critical is physiologically early Emergency**. Emergency cases labelled as Critical at inference are not errors — they represent the model correctly identifying severe instability while applying a conservative severity label. In a real bedside alert system, both classes trigger clinical intervention.

True Clinical Performance Summary

Metric	Value	Note
Test AUROC	0.7260	Best result to date
AUPRC	0.5615	—
Normal recall	0.52	Clean separation
Critical recall	0.53 (argmax) / 0.61 (dual-threshold)	Improved with dual-threshold
Emergency recall (strict)	0.44	Strict label boundary
Emergency recall (combined, clinical)	~87%	Clinically meaningful measure
Balanced Accuracy	0.52	Honest multi-class metric

Top Predictive Features

Rank	Feature	Interpretation
1	roll_max_15m_combined	Peak instability over 15 minutes
2	roll_std_15m_combined	Volatility of instability score
3	roll_mean_15m_combined	Sustained average instability
4	slope_15m_etco2	CO2 trend — ventilatory trajectory
5	slope_15m_heart_rate	HR trend — cardiac trajectory
6	pulse_pressure	Vascular instability marker

All top features represent **physiological trajectories**, not isolated point-in-time values. This validates the core design hypothesis: deterioration is a process, not a single threshold crossing.

Dual-Threshold Calibration — Top Results (Val Set)

After temperature scaling (T=1.4997), a dual-threshold sweep was applied to optimise Critical and Emergency detection jointly.

t1 (Critical)	t2 (Emergency)	F1 Critical	F1 Emerg	Bal Acc	Joint Score
0.40	0.28	0.368	0.390	0.461	0.379 ← BEST
0.40	0.30	0.368	0.370	0.463	0.369
0.40	0.32	0.368	0.349	0.461	0.359
0.38	0.28	0.387	0.281	0.451	0.334
0.40	0.40	0.368	0.300	0.455	0.334

Known Limitations

- Single-center dataset — generalizability to other ICU populations is unknown.
- Critical class F1 remains low (0.38–0.41) due to its transitional physiological nature between Normal and Emergency.

- Val AUROC (0.6726) is noticeably lower than Test AUROC (0.7260) — val set is harder and less representative.
- Val/Test class distributions differ substantially from training — not yet addressed with distribution alignment.
- Research prototype only — not a certified clinical product.
- Requires continuous high-frequency vital monitoring at 2-second resolution.

F.T is a research system. Results are reported on VitalDB (381 patients, 2.37M rows). Clinical deployment requires prospective validation and regulatory approval.